

TMS

Parent Portal and Teacher Portal

Complete functional and technical documentation

This guide explains what every section does, how users operate it, how data is loaded or changed, which validations apply, and how actions flow between parents, teachers, students, Branch Admins, and Tenant Admins.

Audience	Parents, teachers, administrators, support staff, QA teams, and developers
Implementation scope	Current React web portals and Express/Prisma backend in this repository
Security model	Authenticated tenant isolation, linked-child scoping, assigned-class ownership, and branch permissions

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How to read this guide

Each portal section uses the same five-part description: purpose, user workflow, system behavior, rules, and cross-portal effect. API paths are included for support and development teams. A section marked as read-only never mutates data from that screen.

1. System scope and shared behavior

The two portals are different views over the same tenant-scoped institutional records. They are not separate databases. A teacher action can immediately change what a linked parent sees after refresh, provided the record is published and passes visibility rules.

Authentication and isolation

- Every portal request requires an authenticated session and an active user account.
- The tenant identifier comes from the signed session; clients cannot choose another institution.
- Parents can query only students explicitly linked through the parent-student relationship.
- Teachers can mutate only classes assigned through the class teacherId relationship within their tenant.
- Branch-scoped administrative actions resolve and verify the exact assigned branch.

Shared refresh model

Both portals load a consolidated workspace dataset. After a successful mutation - for example sending a message, submitting leave, saving attendance, or publishing results - the portal reloads the dataset so lists, counters, and statuses reflect server state.

Status and visibility principles

Draft vs published	Result drafts remain private. Published results become visible to student/parent-facing reporting and are locked from ordinary teacher deletion.
Parent-visible remarks	Only remarks with parentVisible=true are returned. The API also derives performance signals from published scores.
Blocked enrollment	A blocked student remains visible, but cannot be marked present. Fee arrears can also produce a blocked warning in the Parent Portal.
Approved leave	Approved student leave is represented in attendance as Absent (Excused), not Present.
Child/class context	Parent screens follow the selected child. Teacher screens follow assigned classes and selected result events.

2. Parent Portal

The Parent Portal is a private, child-specific workspace. If a parent has multiple linked children, the child switcher reloads the complete dataset using that student's ID. The URL keeps the selected child in the query string so navigation stays scoped.

Common parent workflow

- Choose a linked child. Every subsequent record belongs only to that child.
- Use the page navigation to open schedules, attendance, communication, fees, and documents.
- Read-only records come from institution workflows; permitted forms submit validated requests.
- Successful submissions refresh the selected child's dataset without mixing sibling records.

2.1 Family overview (Home)

Purpose	Provides today's priorities and summary indicators for the selected child.
How the user works with it	1. Select the child from linked students. 2. Review fee-block warnings, attendance percentage, today's sessions, remarks, events, and counters. 3. Use quick links to open the detailed section.
How the system works	The portal loads /parent/portal for the selected student, calculates present attendance rate, outstanding unpaid/overdue invoices, today's schedule slots, recent visible remarks, appointments, leave records, and upcoming events.
Rules and validation	Records are child-scoped. A linked child with blocked enrollment or overdue invoices shows an alert, but historical records remain visible.
Cross-portal effect	Summarizes teacher attendance/results/remarks and administrator billing/calendar work.

Primary integration: GET /api/parent/portal?studentId=...

2.2 Timetable

Purpose	Shows today's merged schedule across all current regular and activity enrollments for one child.
How the user works with it	1. Open Timetable. 2. Read start/end time, subject, assigned teacher, room or class name, and course type. 3. Switch children to reload a different schedule.
How the system works	The API filters active, non-expired enrollments, normalizes each class schedule, selects slots matching today's Kathmandu weekday, and sorts them by start time.
Rules and validation	Sibling schedules are never merged. Unassigned teachers and missing rooms receive explicit fallback labels.
Cross-portal effect	Branch/Tenant timetable changes and teacher assignment changes appear here after refresh.

Primary integration: GET /api/parent/portal

Current implementation note: This view currently presents today's matching slots rather than a seven-day calendar.

2.3 Attendance

Purpose	Provides read-only session attendance and the child's overall present rate.
How the user works with it	1. Open Attendance. 2. Review the percentage ring. 3. Inspect date, subject, class/session, and Present, Absent, or Absent (Excused) status.

How the system works	Up to 60 recent StudentAttendance records are returned. PRESENT counts toward the displayed rate; EXCUSED maps to Absent (Excused).
Rules and validation	Parents cannot edit attendance. Teacher presence is required before class attendance can be recorded; blocked students cannot be marked present.
Cross-portal effect	Teacher attendance submissions populate this screen. Approved leave can convert covered absence to excused status.

Primary integration: GET /api/parent/portal

2.4 Performance and remarks

Purpose	Displays approved observations and score-derived performance signals without exposing internal notes.
How the user works with it	1. Open Performance and remarks. 2. Review subject, signal, narrative, author, and date. 3. Use Improving, Stable, or Needs support as guidance rather than a final grade.
How the system works	The backend returns only parentVisible remarks. It groups published scores by subject, calculates percentages and average, and compares latest vs previous to derive a signal.
Rules and validation	Unpublished scores and internal remarks are excluded. No editing is available to parents.
Cross-portal effect	Published teacher/admin results and approved staff remarks become parent insights.

Primary integration: GET /api/parent/portal

Current implementation note: This screen summarizes score trends; individual mark-sheet viewing belongs to result/student reporting workflows.

2.5 Teacher messages

Purpose	Supports private, child-and-recipient-specific conversations with assigned teachers and eligible Branch Admin contacts.
How the user works with it	1. Choose a contact. 2. Read the thread for the selected child and contact. 3. Write up to 4,000 characters and send.
How the system works	The API verifies that the parent is linked to the student and that the receiver is an assigned teacher or permitted branch contact, then persists ParentMessage and reloads the thread.
Rules and validation	Threads are separated by student and participant. A parent cannot message arbitrary institution users.
Cross-portal effect	Teachers/admins receive a persisted message and can respond through authorized communication workflows.

Primary integration: POST /api/communication/messages

2.6 Appointments

Purpose	Lets a parent request a student-related meeting with the assigned Branch Admin and track decisions.
How the user works with it	1. Choose a future date/time beyond the configured booking window. 2. Enter the reason. 3. Submit and track Requested, Approved, Rejected, Alternative proposed, or Confirmed.
How the system works	The tenant's appointmentWindowHours controls the minimum lead time. The server verifies child linkage, branch scope, target Branch Admin, and persists response/alternative metadata.
Rules and validation	A branch assignment is mandatory. Client and server both enforce advance notice and valid participant scope.
Cross-portal effect	Branch Admin responses update the parent's appointment history and notification feed.

Primary integration: POST /api/appointments/request

2.7 Leave management

Purpose	Submits planned student leave and displays planned leave plus emergency departure history.
How the user works with it	1. Select start/end dates. 2. Choose Medical, Family event, Travel, or Other. 3. Provide details and submit. 4. Track Pending, Approved, Rejected, or Emergency departure.
How the system works	Medical maps to SICK; other choices map to CASUAL. The server validates child, branch, dates, approval path, and notifies relevant branch administrators/teachers.
Rules and validation	End date cannot precede start date. Only linked children in an active branch can be submitted.
Cross-portal effect	Approved leave affects attendance as Absent (Excused). Branch-recorded emergency departure appears in history and notifications.

Primary integration: POST /api/leaves/request

2.8 Fees and payment

Purpose	Explains admission and monthly billing, shows invoice history, renders official bills, and produces Nepal Pay QR data for unpaid invoices.
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How the user works with it	1. Review one-time admission status and enrollment validity. 2. Review monthly billing mode and recurring line items. 3. Select an invoice to view its itemized bill. 4. For unpaid invoices, open the Nepal Pay QR and verify child, merchant, reference, and amount before payment.
How the system works	The backend calculates billing from grade/subject mode and active activities, returns invoice snapshots, discount/fine lines, due-state classification, and an authenticated invoice-specific Nepal Pay payload.
Rules and validation	Paid invoices do not expose a payment QR. Invoices are child-specific. Overdue invoices contribute to blocked status and outstanding total.
Cross-portal effect	Finance invoice generation and payment confirmation change status, enrollment validity, alerts, and access-related indicators.

Primary integration: [GET /api/parent/portal](#); [GET /api/finances/nepalpay-qr/:invoiceId](#)

Current implementation note: The QR screen generates payment instructions; final payment status depends on the configured gateway/reconciliation workflow.

2.9 Certificates

Purpose	Maintains a permanent downloadable history of certificates issued to the child.
How the user works with it	1. Open Certificates. 2. Review title, grade/course context, issued date, and certificate ID. 3. Download the PDF when available.
How the system works	The API reads issued Certificate records and maps each to an authenticated certificate download endpoint.
Rules and validation	Only certificates belonging to the linked child are visible. Parents cannot issue or alter certificates.
Cross-portal effect	Branch/Tenant certificate issuance immediately adds the document and a notification.

Primary integration: [GET /api/certificates/:certificateId/download](#)

2.10 Academic calendar

Purpose	Shows institution-wide and child-branch events such as holidays, exams, ceremonies, and fee deadlines.
How the user works with it	1. Open Academic calendar. 2. Review event type, title, date, and details. 3. Switch child to change branch-scoped dates.
How the system works	The API combines tenant-wide events with events for branches represented by the child's current enrollments, then orders them by start date.
Rules and validation	Events from unrelated branches and siblings are excluded.

Cross-portal effect

Administrator calendar publishing feeds both Parent and Teacher operational planning.

Primary integration: GET /api/parent/portal

2.11 Notifications

Purpose	Aggregates actionable fee, attendance, emergency, appointment, and certificate activity.
How the user works with it	1. Open a notification to mark it read and navigate to its destination. 2. Use Mark all as read when appropriate. 3. Check Push/SMS channel labels and urgent styling.
How the system works	Notifications are derived from current records and sorted by occurrence time. Read IDs are stored locally per child in the browser.
Rules and validation	Read state is device/browser-local. Urgent events include overdue fees and emergency departures. SMS is reserved for selected high-value events.
Cross-portal effect	Actions across finance, attendance, leave, appointments, and certificates generate the feed inputs.

Primary integration: GET /api/parent/portal

Current implementation note: Read/unread persistence currently uses localStorage, not a server-side notification receipt table.

2.12 Security

Purpose	Allows the signed-in parent to change the account password.
How the user works with it	1. Open Security. 2. Enter the current password and a valid new password. 3. Submit and respond to validation or authentication feedback.
How the system works	The shared ChangePasswordForm calls the authentication service. Password rules and current-password verification are enforced centrally.
Rules and validation	The action affects the parent account, not the selected child. Tenant data is unchanged.
Cross-portal effect	Protects access to all linked-child records.

Primary integration: Authentication change-password endpoint

3. Teacher Portal

The Teacher Portal is an assigned-class workspace. One consolidated `/teacher/workspace` response supplies profile, class rosters, schedules, sessions, syllabi, homework, result events, attendance stamps, leave history, and payroll data.

Common teacher workflow

- The teacher signs in and receives only classes whose `teacherId` matches the authenticated user in the current tenant.
- Class pickers constrain mutations to owned classes and enrolled students.
- Successful mutations reload the workspace so student/parent/admin views can reflect the change.
- Draft/publish, attendance presence, geofence, and daily-update gates prevent incomplete or unauthorized operations.

3.1 Teacher dashboard

Purpose	Summarizes attendance, completed sessions, update compliance, assigned classes, today's timetable, and pending work.
How the user works with it	1. Review attendance days, session count, compliance percentage, and class count. 2. Open pending daily updates. 3. Use quick actions for attendance, syllabus, homework, or results.
How the system works	The workspace aggregates teacher stamps, today's TeacherSession rows, the latest class sessions, assigned classes, and dailyUpdateSubmitted states.
Rules and validation	Statistics are tenant- and teacher-scoped. Pending updates remain visible until submitted.
Cross-portal effect	Provides entry points to records consumed by students, parents, and Branch Admin workflows.

Primary integration: GET /api/teacher/workspace

3.2 My timetable

Purpose	Shows today's sessions and the recurring seven-day schedule for every assigned class.
How the user works with it	1. Review today's scheduled sessions and update status. 2. Scan the current-week grid by weekday. 3. Check class, subject, room, branch, and time.
How the system works	Class schedules are normalized from stored JSON. Today's TeacherSession records are displayed separately from recurring weekly slots.
Rules and validation	The teacher cannot edit the timetable here; Branch/Tenant Admins own class scheduling and conflict checks.
Cross-portal effect	Administrative timetable changes flow to this view and to enrolled child schedules.

Primary integration: GET /api/teacher/workspace

3.3 Geo attendance

Purpose	Records server-verified teacher IN/OUT stamps at an assigned branch.
How the user works with it	1. Choose an assigned branch. 2. Allow precise browser location. 3. Press Mark IN or Mark OUT. 4. Review GPS accuracy and recent stamps.
How the system works	The browser requests a fresh high-accuracy position. The API verifies permission, pending-update gate, GPS accuracy (maximum 20 m), tenant branch, distance from branch center, and allowed radius before saving TeacherAttendance.

Rules and validation	IN is disabled while already checked in; OUT is disabled while checked out. Location denial, timeout, poor accuracy, and geofence violations are surfaced.
Cross-portal effect	Presence gates class attendance and feeds profile/dashboard attendance metrics and Branch Admin visibility.

Primary integration: POST /api/attendance/in; POST /api/attendance/out

Current implementation note: Automatic OUT/RE-IN policy depends on operational attendance automation beyond the manual controls.

3.4 Class attendance

Purpose	Marks today's attendance for students in an assigned class and displays cumulative record counts.
How the user works with it	1. Select an assigned class. 2. Review the roster. 3. Choose Present or Absent for each student. 4. Save today's attendance.
How the system works	The server confirms teacher class ownership, today's date, teacher presence, valid roster membership, and creates/updates the day's TeacherSession and StudentAttendance records.
Rules and validation	Only today's date is accepted. Blocked enrollments cannot be Present. Approved leave can result in EXCUSED status.
Cross-portal effect	Parents and Branch Admins see submitted attendance; teacher dashboard/profile metrics update.

Primary integration: POST /api/teacher/class/:classId/attendance

3.5 Syllabus tracker

Purpose	Creates and maintains a class syllabus with ordered chapters and detailed topics.
How the user works with it	1. Select a class. 2. Create or edit subject and chapter titles. 3. Add, rename, or delete topics within chapters. 4. Save changes.
How the system works	The API validates class ownership and syllabus authorship, persists ordered chapters/topics, and protects topics with progress history from unsafe deletion.
Rules and validation	At least one chapter is required. All mutations are limited to the assigned class and creating teacher.
Cross-portal effect	Syllabus content and progress are visible to Branch Admin monitoring and student learning views.

Primary integration: POST/PATCH /api/teacher/syllabus; topic endpoints

3.6 Daily class update

Purpose	Records chapter/topic progress and confirms completed teaching sessions with a lesson summary.
How the user works with it	1. Select a class and syllabus. 2. Add an optional note and set topic/chapter to Untouched, In progress, or Completed. 3. For pending sessions, enter the required class summary and submit.
How the system works	Progress uses date-based upserts for DailyLessonLog and TopicProgressLog. Topic updates also recalculate parent chapter state. Session submission sets dailyUpdateSubmitted=true and status=PRESENT_CONFIRMED.
Rules and validation	A session summary cannot be empty. A session can be confirmed once. Prior pending updates can block future geo Mark IN.
Cross-portal effect	Progress appears in Branch Admin syllabus monitoring and student views; confirmed sessions improve compliance metrics.

Primary integration: POST /api/teacher/syllabus/:id/log; /topic-log; /session/:id/update

3.7 Homework

Purpose	Assigns one homework item to one or more assigned classes and lists recent assignments.
How the user works with it	1. Select one or more classes. 2. Enter title, instructions, and due date. 3. Optionally attach a supported file under 120 KB. 4. Submit and review recent homework.
How the system works	The client submits one create request per selected class. The server validates class ownership, deadline, content, and persists Homework linked to the class and creator.
Rules and validation	At least one class and a future/valid due date are required. The current client encodes small attachments as data URLs.
Cross-portal effect	Students and Branch Admin monitoring receive the assignment after refresh.

Primary integration: POST /api/homework

Current implementation note: Durable object storage is not yet used for teacher attachments; the current small data-URL mechanism is not ideal for production-scale files.

3.8 Results

Purpose	Lets teachers enter marks only for result events created and opened by Branch Admins, save private drafts, publish results, and delete unpublished drafts.
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How the user works with it	1. Select an available result event. 2. Enter full marks and pass marks. 3. Enter a valid mark for every enrolled student and attach each result sheet. 4. Save draft or Publish now. 5. Review saved results; delete only unpublished drafts.
How the system works	The workspace returns open ResultDefinition records for assigned classes and score history for those events. The server validates event/class/title/subject, roster coverage, limits, attachments, computes percentile, and upserts one StudentScore per event/student. Publishing stamps publishedAt.
Rules and validation	Marks must be 0..maximum; pass marks 0..maximum. Every current student needs a mark and sheet. Closed events are unavailable. Published scores are locked from teacher deletion.
Cross-portal effect	Branch Admin result setup enables teacher entry. Published scores feed student/parent performance visibility and admin reports.

Primary integration: POST /api/teacher/results/:results/share; DELETE /results/:id

Current implementation note: Result sheets currently use small data URLs rather than production object storage.

3.9 My profile

Purpose	Shows institutional identity, employment details, performance summary, and attendance stamp history.
How the user works with it	1. Review name, email, designation, contract, joining date, and assigned branches. 2. Review attendance and update-compliance percentages. 3. Inspect IN, OUT, AUTO-OUT, and RE-IN history.
How the system works	Profile combines the authenticated user, StaffRecord, assigned classes/branches, performance score, computed metrics, and recent TeacherAttendance rows.
Rules and validation	The page is read-only. Institutional record corrections require an administrator.
Cross-portal effect	Uses data produced by HR, timetable assignment, geo attendance, and daily updates.

Primary integration: GET /api/teacher/workspace

3.10 Leave requests

Purpose	Submits teacher leave through the configured branch/tenant approval path and shows status history.
How the user works with it	1. Choose leave type. 2. Enter start/end dates and reason. 3. Submit. 4. Track pending branch review, tenant review, approval, or rejection.

How the system works	The request is linked to the teacher's branch. CASUAL, SICK, EARLY_OUT, and LONG_SICK follow configured approval rules; LONG_SICK can require two levels.
Rules and validation	A branch assignment is required. End date cannot precede start date. Reasons are length-limited.
Cross-portal effect	Branch/Tenant approval changes history and teacher attendance reporting.

Primary integration: POST /api/leaves/request

3.11 Salary slips

Purpose	Shows issued payroll history and a one-month projection based on the latest payroll values.
How the user works with it	1. Open Salary slips. 2. Choose a period. 3. Review base salary, attendance deductions, bonuses, net payable, status, and paid date. 4. Treat projected values as estimates.
How the system works	The workspace reads StaffRecord payrolls ordered newest first. The UI creates a client-side next-month projection by copying the latest issued amounts.
Rules and validation	Teachers cannot edit payroll. Projection is informational and may change after attendance/admin adjustments.
Cross-portal effect	HR/finance payroll issuance and settlement populate official rows.

Primary integration: GET /api/teacher/workspace

Current implementation note: The projected slip is not a persisted or approved payroll record.

3.12 Security

Purpose	Allows the signed-in teacher to change the account password.
How the user works with it	1. Open Security. 2. Enter current and new password values. 3. Submit and resolve any validation errors.
How the system works	The shared authentication form verifies the current password and applies central password policy.
Rules and validation	This changes only account credentials and does not alter teacher records or classes.
Cross-portal effect	Protects all teacher-class and student data accessible through the account.

Primary integration: Authentication change-password endpoint

3.13 Shared teacher data lifecycle

Purpose	Explains how teacher actions progress from editable work to institution-visible records.
How the user works with it	1. Work inside an assigned class. 2. Save validated records. 3. Publish or confirm when the workflow requires it. 4. Reload to verify downstream visibility.
How the system works	Ownership checks use teacherId, tenantId, classId, and creator/recorder IDs. ResultDefinition controls results; TeacherSession controls attendance/update compliance; Syllabus/Homework link directly to class.
Rules and validation	No teacher mutation should accept arbitrary tenant, branch, class, student, or result ownership supplied by the client.
Cross-portal effect	Confirmed/published records become available to parents, students, Branch Admins, finance/HR summaries, or notifications according to visibility rules.

Primary integration: GET/POST/PATCH/DELETE teacher and supporting routes

4. Cross-portal workflows

4.1 Timetable assignment

Branch/Tenant Admin creates a class timetable and assigns a teacher. The same normalized schedule feeds Teacher My timetable and the selected child's Parent Timetable. Conflict checks protect teacher, room, and overlapping enrolled-student availability before save.

4.2 Attendance and leave

Teacher marks IN inside the branch geofence, then records today's class roster. Planned leave is submitted by a parent and approved by administrators. Covered attendance can be stored/displayed as Absent (Excused). Parent Attendance is read-only and receives the resulting StudentAttendance records.

4.3 Syllabus and daily progress

Teacher creates syllabus/chapter/topic structure and posts dated progress. Branch Admin sees live progress for oversight; student-facing learning views receive the shared curriculum state. Daily summaries also confirm sessions and improve teacher compliance metrics.

4.4 Homework

Teacher assigns homework to owned classes. Branch Admin can monitor branch homework, and enrolled students receive the class assignment. Parent Portal does not currently expose a dedicated homework section.

4.5 Results

Branch Admin creates an open result event for an assigned class. Teacher enters/upserts student drafts, then publishes. Published scores become eligible for student reporting and parent score-derived performance signals. Drafts remain private.

4.6 Communication and appointments

Parent communication is constrained by linked child and assigned/eligible recipient. Appointment requests target the child's Branch Admin and respect tenant lead-time policy. Responses and alternatives return to the Parent Portal history and notifications.

4.7 Billing and blocked status

Finance creates admission and recurring invoices. Unpaid/overdue amounts appear in Parent Fees; overdue status can show a blocked warning. A blocked enrollment remains visible but teacher attendance cannot mark the student Present.

5. Operational notes and known limitations

- Portal data freshness: successful mutations trigger reloads; external changes require refresh or re-entry to the section.
- Teacher result and homework attachments currently use small browser data URLs. Production deployment should use authenticated object storage, scanning, size/type policy, and expiring download URLs.
- Parent notification read state is browser-local. Cross-device read synchronization requires server-side notification receipts.
- Parent Timetable currently maps today's weekly slots rather than rendering a full seven-day grid.
- Teacher salary projection is a UI estimate copied from the latest payroll, not an approved accounting record.
- Permissions must remain tenant- and branch-scoped. Resource-ID endpoints should resolve ownership before mutation rather than trusting client-supplied scope.

Support checklist

Portal does not load	Verify authenticated active account, tenant scope, linked parent/student or teacher class assignment, API health, and database migrations.
Parent sees no child	Create/verify Parent and StudentParent linkage in the same tenant.
Teacher sees no class	Assign the teacher's user ID to Class.teacherId and verify course tenant/branch.
Teacher sees no result	Verify an open ResultDefinition exists for a class assigned to that teacher.
Attendance save fails	Verify teacher is marked IN today, class is assigned, student is enrolled, and blocked students are not Present.
Geo attendance fails	Check browser permission, secure context, GPS accuracy ≤ 20 m, assigned branch coordinates/radius, and pending daily updates.
Parent payment unavailable	Verify invoice ownership, unpaid status, gateway configuration, and authenticated QR endpoint access.

End of guide

This document describes the repository implementation as reviewed on 2 September 2026. Update it when routes, permissions, status models, storage, or portal navigation change.